

Raj Vaghela

AI Systems Engineer | LLM Applications | Data Pipelines | Full Stack Development

Leicester, UK | vaghela.raj2581@gmail.com

rajvaghela.dev | github.com/Raj-Vaghela | linkedin.com/in/raj-vaghela

Profile

AI Systems Engineer building LLM applications, data pipelines and full stack tools at Stack8s. Own pricing workflows across 22 cloud providers, an agentic recommendation assistant and Supabase operations. MSc Advanced Computer Science graduate with Distinction, with hands-on experience in Python, FastAPI, Next.js, PostgreSQL and retrieval-augmented generation.

Experience

Stack8s | AI Systems Engineer

Jan 2026 to present | Remote, UK

- Build and maintain automated pricing pipelines across 22 cloud providers, with daily ingestion, standardisation and validation for product and analytics use.
- Develop an agentic assistant that recommends compute instances, AI models and application stacks, owning implementation from backend to interface.
- Ingest and structure AI model metadata from Hugging Face and NVIDIA NIMs to support model discovery and comparison.
- Build a GPU availability tracker that checks providers approximately every three minutes.
- Create internal dashboards for analytics, pricing and margin changes, including controlled database writeback workflows.
- Own Supabase Auth, PostgreSQL databases and Storage; troubleshoot application, data and infrastructure issues through delivery.

Direction Infosystems | Web Development Intern

Jan 2023 to Apr 2023 | Bardoli, India

- Built and maintained websites using PHP, Laravel, MySQL, Bootstrap and jQuery. Assisted with interface design, deployment and handover in a team environment.

Technical skills

Languages and web: Python, TypeScript, JavaScript, SQL, Next.js, React, FastAPI, Node.js, Express, REST APIs

Applied AI: RAG, embeddings, pgvector, tool/function calling, agent workflows, Gemini, OpenAI, Hugging Face, NVIDIA NIMs, Whisper, TTS, OCR

Data and platforms: ETL, pandas, NumPy, PostgreSQL, Supabase Auth and Storage, schema design, background jobs, Vercel, Azure, Google Cloud, Git/GitHub

Engineering workflow: CI/CD, logging, retries/backoff, debugging, code review; Cursor, Codex and Claude Code

Selected projects

Medical Screening Assistant

MSc dissertation | 2025 | Next.js, FastAPI, Gemini, pgvector

- Built a nurse-supervised screening prototype with RAG over NHS and MedlinePlus content and hospital-specific context for symptom intake and department suggestions.
- Integrated Whisper STT, ElevenLabs TTS and OCR into the interface; implemented document retrieval and source citations.
- Added staff controls for hospital policies and role-based access. Nurses retain authority over suggestions.

Job Recruiter Assistant

Personal project | 2025 | Next.js, FastAPI, Supabase, Gemini

- Designed a five-table PostgreSQL schema and database function for semantic candidate matching with evidence-linked answers.
- Built background CV processing with validation, storage, OCR and embeddings, keeping document ingestion separate from chat.
- Implemented draft, preview and approval steps for SendGrid outreach, with per-recipient delivery results.

Crypto FM

Encode AI London hackathon | 2025 | Node.js, Gemini, Google Cloud TTS

- Connected market data collectors, analysis and radio script agents to a text-to-speech pipeline for narrated broadcasts.
- Implemented a segment queue, quota-aware collection, retry/backoff handling and structured logs for continuous playback.

30-Day Readmission Prediction

Machine learning project | 2024 | Python, pandas, scikit-learn

- Built a reproducible pipeline on 100,000+ hospital encounters, handling class imbalance and exploring patient segmentation for readmission classification.

Education

MSc Advanced Computer Science, Distinction

University of Leicester | Jan 2024 to Jul 2025

BE Computer Science and Engineering, 8.55/10

S.N. Patel Institute of Technology and Research Centre | Jun 2019 to Jun 2023

Earlier internship and training

Elsner Technologies summer internship, Jun to Jul 2022: built an Android to-do app with Java. IBM virtual internship programme, Jun to Jul 2023: Python and machine learning exercises. Microsoft Future Ready Talent virtual internship, Apr to Jun 2023: Azure hosting and GitHub CI/CD.

Highlights

Huawei Tech Arena finalist, top 8 of 100+ teams; led a four-person team to a proof of concept and board pitch. University peer mentor, course representative and Leicester 100 participant. MYSY Merit Scholarship recipient.